

BdMO 2026 National Basic Inequalities Problem Set

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§1 AM-GM Inequality

Theorem (AM-GM Inequality)

For any real numbers $x_1, x_2, \dots, x_n \geq 0$,

$$\frac{x_1 + x_2 + \dots + x_n}{n} \geq \sqrt[n]{x_1 x_2 \dots x_n}$$

with equality if and only if $x_1 = x_2 = \dots = x_n$.

Problem 1.1. Let a, b, c be positive reals. Prove that,

$$a^2 + b^2 + c^2 \geq ab + bc + ca.$$

Problem 1.2. Let $a_1, a_2, \dots, a_n$ be positive real numbers such that $a_1 a_2 \dots a_n = 1$. Prove that

$$(1 + a_1)(1 + a_2) \dots (1 + a_n) \geq 2^n.$$

Problem 1.3. Let a, b, c be nonnegative real numbers. Prove that

$$(a + b)(b + c)(c + a) \geq 8abc.$$

Problem 1.4. Let $a, b, c > 0$. Prove that

$$\frac{a^3}{bc} + \frac{b^3}{ca} + \frac{c^3}{ab} \geq a + b + c.$$

§1.1 AM-GM Buffer Problems

Problem 1.5. Let a, b, c be positive reals such that $a^4 + b^4 + c^4 = 8$ and $a^2 + b^2 + c^2 = 67$. Prove that

$$a^3 + b^3 + c^3 \leq 69.$$

Problem 1.6. Let $n \geq 3$ be an integer, and let $a_2, a_3, \dots, a_n$ be positive real numbers such that $a_2 a_3 \dots a_n = 1$. Prove that

$$(1 + a_2)^2 (1 + a_3)^3 \dots (1 + a_n)^n > n^n.$$

Problem 1.7. Prove that for all a, b, c positive real numbers

$$\frac{a^4 + 1}{b^3 + b^2 + b} + \frac{b^4 + 1}{c^3 + c^2 + c} + \frac{c^4 + 1}{a^3 + a^2 + a} \geq 2.$$

Problem 1.8. Suppose that a, b, c, d are positive real numbers satisfying $(a + c)(b + d) = ac + bd$. Find the smallest possible value of

$$\frac{a}{b} + \frac{b}{c} + \frac{c}{d} + \frac{d}{a}.$$

§2 Cauchy-Schwarz Inequality

Theorem (Cauchy-Schwarz)

For any list of reals $a_1, a_2, \dots, a_n$ and $b_1, b_2, \dots, b_n$,

$$(a_1^2 + a_2^2 + \dots + a_n^2)(b_1^2 + b_2^2 + \dots + b_n^2) \geq (a_1b_1 + a_2b_2 + \dots + a_nb_n)^2$$

with equality if and only if there exists a constant t such that $a_i = tb_i$ for all $1 \leq i \leq n$, or if one list consists of only zeroes.

Problem 2.1. For positive reals a, b, c . Prove that

$$(a + b + c) \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) \geq 9.$$

Problem 2.2. For positive reals a, b, c . Prove that

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2}$$

Problem 2.3. If P is a polynomial with positive real coefficients. Prove that for all x and y in positive reals that

$$P(x^2)P(y^2) \geq P(xy)^2$$

§2.1 Titu's Lemma/Engel's Form/Sedrakyan's Inequality

Theorem (Titu's Lemma)

Titu's lemma states that:

$$\frac{a_1^2}{b_1} + \frac{a_2^2}{b_2} + \dots + \frac{a_n^2}{b_n} \geq \frac{(a_1 + a_2 + \dots + a_n)^2}{b_1 + b_2 + \dots + b_n}.$$

It is a direct consequence of Cauchy-Schwarz inequality.

Problem 2.4. Given that positive reals a, b , and c are subject to $a + b + c = 1$, find the minimum value of

$$\frac{1}{a} + \frac{4}{b} + \frac{9}{c}$$

Problem 2.5. Let a, b, c be positive real numbers such that $abc = 1$. Prove that

$$\frac{1}{a^3(b+c)} + \frac{1}{b^3(c+a)} + \frac{1}{c^3(a+b)} \geq \frac{3}{2}.$$

§2.2 Cauchy/Titu Buffer Problems

Problem 2.6. Let x, y, z be real numbers greater than 1. Prove that if $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 2$ then

$$\sqrt{x+y+z} \geq \sqrt{x-1} + \sqrt{y-1} + \sqrt{z-1}.$$

Problem 2.7. Let a, b, c, x, y be positive real numbers such that $abc = 1$. Prove that

$$\frac{a^5}{xc+yb} + \frac{b^5}{xa+yc} + \frac{c^5}{xb+ya} \geq \frac{9}{(x+y)(a^2+b^2+c^2)}.$$

§3 Hölder's Inequality

Theorem (Hölder's Inequality)

Let p and q be positive real numbers. Let $a_1, \dots, a_n$ and $b_1, \dots, b_n$ be nonnegative real numbers. Then

$$\left(\sum_{i=1}^n a_i \right)^p \left(\sum_{i=1}^n b_i \right)^q \geq \left(\sum_{i=1}^n \sqrt[p+q]{a_i^p b_i^q} \right)^{p+q}.$$

Problem 3.1. Let a, b, c be positive reals. Prove that

$$\frac{a}{\sqrt{a^2+8bc}} + \frac{b}{\sqrt{b^2+8ca}} + \frac{c}{\sqrt{c^2+8ab}} \geq 1$$

Problem 3.2. For $a, b, c > 0$ prove that

$$\frac{a}{\sqrt{b+c}} + \frac{b}{\sqrt{c+a}} + \frac{c}{\sqrt{a+b}} \geq \sqrt{\frac{3}{2}}(a+b+c).$$

§3.1 Hölder Buffer Problems

Problem 3.3. Let x, y, z be positive real numbers which satisfy $xyz = 1$. Show that

$$\sum_{\text{cyc}} \frac{x^3}{(1+y)(1+z)} \geq \frac{3}{4}.$$

Problem 3.4. If a, b, c are three positive real numbers such that $ab+bc+ca=1$, prove that

$$\sqrt[3]{\frac{1}{a}+6b} + \sqrt[3]{\frac{1}{b}+6c} + \sqrt[3]{\frac{1}{c}+6a} \leq \frac{1}{abc}.$$

Problem 3.5. Let all possible 2023-degree real polynomials: $P(x) = x^{2023} + a_1x^{2022} + a_2x^{2021} + \dots + a_{2022}x + a_{2023}$, where $P(0) + P(1) = 0$, and the polynomial has 2023 real roots $r_1, r_2, \dots, r_{2023}$ [not necessarily distinct] so that $0 \leq r_1, r_2, \dots, r_{2023} \leq 1$. What is the maximum value of $r_1 \cdot r_2 \cdot \dots \cdot r_{2023}$?